Department of Computer Science
BAI - 2
Lab Mid (Version 1)
All questions are mapped to CLO4

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A hospital wants to develop a system to manage patient appointments. Each patient has an assigned appointment containing a doctor's name, department, and consultation fee. The hospital manages regular patients and emergency patients. An emergency patient has an additional emergency level and receives priority when appointments are compared.

Implement the following requirements.

Part A – (5 marks)

1. Create an Appointment class containing: doctorName, department, and consultationFee. Include constructors and displayAppointment() to display all appointment details.

2. Create a Patient class containing: patientId, patientName, appointment of type Appointment. Implement constructors that initialize all attributes. The Patient constructor must create its own Appointment object. Include displayDetails() to display the patient's and appointment details.

Part B – (4 marks)

3. Create EmergencyPatient class extends Patient. Add an attribute emergencyLevel (from 1 to 5). Add displayDetails() to display patient and appointment information along with the emergency level.

Part C – (5 marks)

4. In a class named HospitalManager, implement:

   a. void increaseFee(Appointment a): Increase the consultation fee by 10%, but only if the resulting fee does not exceed Rs. 10,000. The fee must be updated in the existing Appointment object, not by creating a replacement object.

   b. void displayPatient(Patient p): display the patient's details by calling displayDetails().

Part D – (6 marks)

5. In HospitalManager, implement: Patient findPatientWithHighestFee(Patient[] patients)
This method must find and return the patient whose appointment has the highest consultation fee. Return null if the array is empty or contains no patient objects. If multiple patients have the same fee, return the first one encountered.

Part E – Main Program and Testing (5 marks)

6. Write the main() method to:

- Create an array capable of storing five Patient objects.
- Store at least three Patient objects and two EmergencyPatient objects.
- Display all patients using the HospitalManager methods.
- Increase the consultation fee of one patient's appointment by passing its existing Appointment object to increaseFee().
- Display that patient again to verify the update.
- Call findPatientWithHighestFee() and display the returned patient's name and consultation fee.

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Lab Mid (Version 2)
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A university wants to develop a Java application to manage students and their assigned final-year projects. Each student is assigned a project containing a project title, supervisor name, and project complexity level. The system must support regular students and research students. A research student has an additional research area and receives a bonus in the evaluation score.

Implement the following requirements.

Part A – (5 marks)

1. Create a Project class containing: projectTitle, supervisorName, and complexityLevel (from 1 to 5). Include a parameterized constructor and a method displayProject() to display all project details.

2. Create a Student class containing: studentId, studentName, and project of type Project. Implement constructors that initialize all attributes. The Student constructor create its own Project object. Include displayDetails() to displays the student's and assigned project details.

Part B – (4 marks)

3. Create a ResearchStudent class that extends Student has an attribute researchArea(string). Initialize attributes and add displayDetails() to display student and project details, including the research area.

Part C – (5 marks)

4. In a class named ProjectManager, implement the following methods:

   a. void increaseComplexity(Project p): Increase the project's complexity level by one, but not exceeds
   5. This must update the existing Project object rather than create a replacement object.

   b. void displayStudent(Student s): Display the student's details by calling its displayDetails().

Part D – (6 marks)

5. In the ProjectManager class, implement: Student findMostComplexProject(Student[] students)
This method must Find and return the student whose assigned project has the highest complexity level. Return null if the array is empty or contains no student objects. If multiple students have the same maximum complexity, return the first one encountered.

Part E – Main Program and Testing (5 marks)

- Create an array capable of storing five Student objects.
- Store at least three Student objects and two ResearchStudent objects in the array.
- Display all students using the ProjectManager methods.
- Increase the complexity of one student's project by passing the existing project object to increaseComplexity().
- Display that student's details again to verify the update.
- Call findMostComplexProject() and display the name of the student with the most complex project.